Math 220: Mathematical Reasoning and Proof
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All Solutions

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Solutions 1: Introduction to course. First proof

Solutions to appear after the assignment is due.

1. Using the definitions of 'divides" and 'even", explain why 0 is even. Explain why 7 divides 0, but 0 does not divide 7.

   Solution. Recall that an integer a is *even* if there exists an integer k such that $a = 2k$. Since $0 = 2 \cdot 0$ and 0 is an integer, it follows that 0 is even.

   Recall also that, for integers a and b, we say that *a divides b* if there exists an integer k such that $b = ak$. Since $0 = 7 \cdot 0$, there exists an integer k, namely $k = 0$, such that $0 = 7k$. Therefore, $7 \mid 0$.

   On the other hand, if 0 divided 7, then there would be an integer k such that $7 = 0 \cdot k$. But $0 \cdot k = 0$ for every integer k, so this would imply that $7 = 0$, which is false. Therefore, $0 \nmid 7$.

2. Suppose that n is an even integer, and let m be any integer. Prove that nm is even.

   Solution. Since n is even, there exists an integer k such that $n = 2k$. Multiplying by m, we obtain

   $$nm = (2k)m = 2(km).$$

   Because k and m are integers, their product km is an integer. Thus nm is twice an integer, and we conclude that nm is even.

3. Suppose that n and m are odd integers. Prove that $n + m$ is an even integer.

   Solution. Since n and m are odd, there exist integers a and b such that $n = 2a + 1$ and $m = 2b + 1$. Therefore,

   $$n + m = (2a + 1) + (2b + 1) = 2a + 2b + 2 = 2(a + b + 1).$$

   Since a and b are integers, $a + b + 1$ is an integer. Thus $n + m$ is twice an integer, and we conclude that $n + m$ is even.

Solutions 2: Logic. "Direct" proofs

1. Write the negation of each of the following statements.

   Solution.

   (a) All triangles are isosceles.
   Negation: There exists a triangle that is not isosceles.

   (b) Every door in the building was locked.
   Negation: There exists a door in the building that was not locked.

   (c) Some even numbers are multiples of three.
   Negation: No even number is a multiple of three.
   Equivalently: Every even number is not a multiple of three.

   (d) Every real number is less than 100.
   Negation: There exists a real number that is greater than or equal to 100.

   (e) Every integer is positive or negative.
   Negation: There exists an integer that is neither positive nor negative.

   (f) If f is a polynomial function, then f is continuous at 0.
   Negation: f is a polynomial function and f is not continuous at 0.

   (g) If $x^2 > 0$, then $x > 0$.
   Negation:
   $$x^2 > 0 \quad \text{and} \quad x \leq 0.$$

   (h) There exists a $y \in \mathbf{R}$ such that $xy = 1$.
   Negation: For every $y \in \mathbf{R}$,
   $$xy \neq 1.$$

   (i) $(2 > 1)$ and $(\forall x, x^2 > 0)$.
   Negation:
   $$(2 \leq 1) \quad \text{or} \quad (\exists x \text{ such that } x^2 \leq 0).$$

   (j) $\forall \epsilon > 0, \exists \delta > 0$ such that if $|x| < \delta$, then $|f(x)| < \epsilon$.
   Negation: There exists an $\epsilon > 0$ such that for every $\delta > 0$, $|x| < \delta$ and $|f(x)| \geq \epsilon$.

   The following is also correct:

   There exists an $\epsilon > 0$ such that for every $\delta > 0$, there exists an x such that
   $$|x| < \delta \qquad \text{and} \qquad |f(x)| \geq \epsilon.$$

2. Write the converse, contrapositive, and negation of each of the following implications.

   Solution.

   (a) If a quadrilateral is a rectangle, then it has two pairs of parallel sides.
   Converse: If a quadrilateral has two pairs of parallel sides, then it is a rectangle.
   Contrapositive: If a quadrilateral does not have two pairs of parallel sides, then it is not a rectangle.
   Negation: There exists a quadrilateral that is a rectangle but does not have two pairs of parallel sides.

(b)

$$(P \wedge \neg Q) \Rightarrow R.$$

Converse:

$$R \Rightarrow (P \wedge \neg Q).$$

Contrapositive:

$$\neg R \Rightarrow \neg(P \wedge \neg Q).$$

Equivalently,

$$\neg R \Rightarrow (\neg P \vee Q).$$

Negation:

$$(P \wedge \neg Q) \wedge \neg R.$$

(c)

$$P \Rightarrow (R \Rightarrow \forall x,\ Q(x)).$$

Converse:

$$(R \Rightarrow \forall x,\ Q(x)) \Rightarrow P.$$

Contrapositive:

$$\neg(R \Rightarrow \forall x,\ Q(x)) \Rightarrow \neg P.$$

Equivalently,

$$(R \wedge \exists x \text{ such that } \neg Q(x)) \Rightarrow \neg P.$$

Negation:

$$P \wedge \neg(R \Rightarrow \forall x,\ Q(x)).$$

Equivalently,

$$P \wedge R \wedge \exists x \text{ such that } \neg Q(x).$$

3. Let P and Q be statements. Write the truth table for

(a) $(\neg P) \vee Q$

(b) $(P \wedge (\neg Q)) \Rightarrow Q$

Solution.

(a)

P	Q	$\neg P$	$(\neg P) \vee Q$
T	T	F	T
T	F	F	F
F	T	T	T
F	F	T	T

(b)

P	Q	$P \wedge \neg Q$	$(P \wedge \neg Q) \Rightarrow Q$
T	T	F	T
T	F	T	F
F	T	F	T
F	F	F	T

4. Are the statements $(P \vee Q) \wedge R$ and $P \vee (Q \wedge R)$ equivalent? If so, give a proof. If not, explain why by giving a counterexample.

   Solution.

   They are not equivalent. For example, suppose $P = T, \quad Q = F, \quad R = F$. Then $(P \vee Q) \wedge R = (T \vee F) \wedge F = T \wedge F = F$, while $P \vee (Q \wedge R) = T \vee (F \wedge F) = T$. Thus the two statements can have different truth values, so they are not equivalent.

5. Let P and Q be statements.

   (a) Prove that $\neg(P \Rightarrow Q)$ is equivalent to $P \wedge \neg Q$.
   Solution.
   We can prove this using a truth table:

P	Q	$\neg(P \Rightarrow Q)$	$P \wedge \neg Q$
T	T	F	F
T	F	T	T
F	T	F	F
F	F	F	F

   The last two columns are identical. Therefore, $\neg(P \Rightarrow Q)$ and $P \wedge \neg Q$ are equivalent.

   (b) Prove that $\neg(P \Rightarrow Q)$ is *not* equivalent to $\neg P \wedge Q$.
   Solution.
   It is enough to find a choice of P and Q for which the two statements have different truth values. Suppose $P = F$ and $Q = T$. Then $P \Rightarrow Q$ is true, so $\neg(P \Rightarrow Q) = F$. On the other hand, $\neg P \wedge Q = T \wedge T = T$. Thus the two statements are not equivalent.

   (c) Give an example of statements P and Q such that $\neg P \Rightarrow \neg Q$ is true and $\neg(P \Rightarrow Q)$ is false.
   Solution.
   For example, let
   $$P = \text{“}2 > 1\text{”} \qquad \text{and} \qquad Q = \text{“}3 > 2\text{”}.$$
   Both P and Q are true.
   Thus $\neg P$ is false, so $\neg P \Rightarrow \neg Q$ is true.
   Also, $P \Rightarrow Q$ is true, and hence $\neg(P \Rightarrow Q)$ is false.

6. Suppose that n is an odd integer. Prove that n^2 is an odd integer.

   Solution.

   Since n is odd, there exists an integer k such that $n = 2k + 1$. Therefore,
   $$\begin{aligned} n^2 &= (2k+1)^2 \\ &= 4k^2 + 4k + 1 \\ &= 2(2k^2 + 2k) + 1. \end{aligned}$$
   Since k is an integer, $2k^2 + 2k$ is an integer. Thus n^2 can be written in the form $2\ell + 1$ for some integer ℓ. Therefore n^2 is odd.

7. Prove that if n^2 is even, then n is even.

   Solution.

   We prove the statement by contraposition.

   The contrapositive is:

If n is not even, then n^2 is not even.

Suppose that n is not even. Since n is an integer, n must therefore be odd. By the previous problem, if n is odd, then n^2 is odd. In particular, n^2 is not even.

Thus the contrapositive is true. Therefore the original statement is true: if n^2 is even, then n is even.

8. Prove that m is even if and only if $m^2 + 6$ is even.

   Solution.

   We prove both directions.

   First suppose that m is even. Then there exists an integer k such that $m = 2k$. Therefore $m^2+6 = (2k)^2+6 = 4k^2 + 6 = 2(2k^2 + 3)$. Since $2k^2 + 3$ is an integer, $m^2 + 6$ is even.

   Conversely, suppose that $m^2 + 6$ is even. Then there exists an integer k such that $m^2 + 6 = 2k$. Then $m^2 = 2k - 6 = 2(k - 6)$. Since $k - 6$ is an integer, m^2 is even. By the previous homework problem, we conclude that m is even.